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Rocket Engine Ignition Structural Shock

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Introduction

An elementary calculation provides the steady state axial load along the rocket body during flight. But, transient loading at ignition is often much larger. Following the sudden increase in thrust at ignition, an elastic compression wave is sent up the structural "stack". This wave produces significant shock loading, especially on fragile payload elements. General experience with large rockets (satellite launch vehicles) shows that this is the primary source of payload (satellite) structural loading.

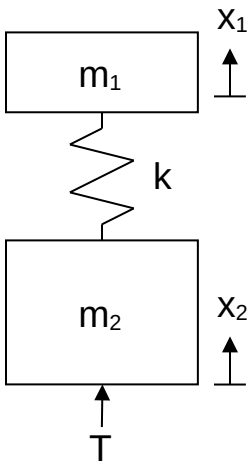
Also, if the thrust vector is misaligned with the forebody center of mass, large bending torques are introduced. Even with small sounding rockets (e.g., NIRO) it is a major loading consideration. More complex models with variable properties along the rocket body, or with solid propellant motors can be readily modeled using finite element techniques.

Nomenclature

Mnemonic	Definition
A	Cross section area of the load-bearing part of the column
Acc	Steady state axial acceleration
$a(x,t)$	Axial acceleration of a column element
E	Young's modulus of the load bearing material
$\sigma(x,t)$	Axial stress, positive in compression
$\varepsilon(x,t)$	Axial strain, positive in tension
L	Column length (or length of a column element)
μ	Column mass per unit length
c	Wave speed of axial elastic displacements
χ	Nominal distance forward of the aft end of the rocket
$\xi(x,t)$	Elastic displacement, positive in the + χ direction
$\xi_{SS}(x)$	Steady State elastic displacement
$\xi_{TR}(x,t)$	Transient vibrational elastic displacement
P	Axial force, positive in tension
s	Laplace transform variable
T	Thrust force
t	Time after rocket ignition
$\bar{b}$	Laplace transform of b(t)

Two Mass Model of Ignition Shock

This memo documents a highly simplified analysis of a rocket structure with a thrust chamber at the extreme aft end. We start by considering a model having two discrete masses connected by a spring:



The two coupled equations of motion are

$$m_2\ddot{x}_2 = T + k(x_1 - x_2) - m_2 g$$
$$m_1\ddot{x}_1 = -k(x_1 - x_2) - m_1 g$$

Here g = acceleration due to gravity.

The initial conditions are

$$\dot{x}_1(0) = 0, \dot{x}_2(0) = 0, x_2(0) = 0, x_1(0) = -\frac{m_1 g}{k}$$

Next, take the Laplace transform (s = the transform variable)

$$m_2 s^2 \bar{x}_2 = \frac{T}{s} + k(\bar{x}_1 - \bar{x}_2) - \frac{m_2 g}{s}$$
$$m_1(s^2\bar{x}_1 + s\frac{m_1 g}{k}) = -k(\bar{x}_1 - \bar{x}_2) - \frac{m_1 g}{s} \quad \text{, or}$$

$$(m_2 s^2 + k)\bar{x}_2 - k\bar{x}_1 = \frac{(T - m_2 g)}{s}$$
$$(m_1 s^2 + k)\bar{x}_1 - kx_2 = -\frac{sm_1^2 g}{k} - \frac{m_1 g}{s} = -\frac{m_1 g}{ks}(m_1 s^2 + k)$$

Then,

$$\bar{x}_2 = (m_1 s^2 + k)(\bar{x}_1 + m_1 g / ks)/k \text{ , and}$$

$$(m_2 s^2 + k)\left[(m_1 s^2 + k)(\bar{x}_1 + \frac{m_1 g}{ks})/k\right] - k\bar{x}_1 = \frac{T - m_2 g}{s}, \text{ or}$$

$$\left[(m_1 m_2 s^4 + k(m_1 + m_2)s^2 + k^2 - k^2\right]\bar{x}_1 + (\frac{m_1 g}{ks})(m_2 s^2 + k) = \frac{k(T - m_2 g)}{s}, \text{ or}$$

$$s^2\left[m_1 m_2 s^2 + k(m_1 + m_2)\right]\bar{x}_1 = \frac{k(T - m_2 g)}{s} - (\frac{m_1 g}{ks})(m_2 s^2 + k).$$

Define the natural frequency of this system to be

$$\omega_n^2 = \frac{k(m_1 + m_2)}{m_1 m_2}.$$

Then,

$$(s^2 + \omega_n^2)\bar{x}_1 = \frac{k(T - m_2 g)}{m_1 m_2 s^3} - (\frac{g}{m_2 k s^3})(m_2 s^2 + k).$$

Now, suppose next that the thrust acceleration is very large compared to g. Then,

$$(s^2 + \omega_n^2)\bar{x}_1 = \frac{kT}{m_1 m_2 s^3}.$$

Given this,

$$\bar{x}_1 = \frac{kT}{m_1 m_2}\frac{1}{s^3(s^2 + \omega_n^2)}.$$

Now, what we really care about is the acceleration, A_1. Then,

$$\bar{A}_1 = \frac{kT}{m_1 m_2}\frac{1}{s(s^2 + \omega_n^2)}.$$

Next, rewrite this in terms of a partial fraction expansion:

$$\frac{m_1 m_2 \bar{A}_1}{kT} = \frac{1}{\omega_n^2 s} - \frac{s}{\omega_n^2(s^2 + \omega_n^2)}$$

The inverse Laplace transform can be found adding the contributions from both terms:

$$\frac{m_1 m_2 \omega_n^2}{kT} A_1(t) = 1 - \cos(\omega_n t).$$

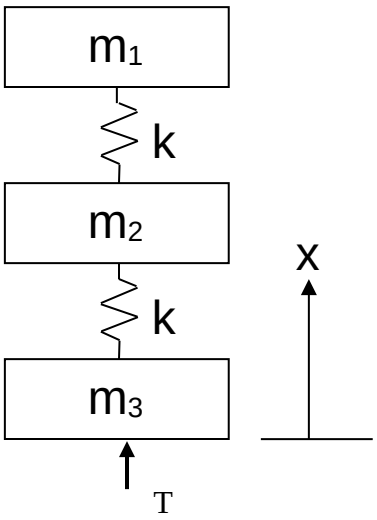
Finally, the maximum value for A_1 is

$$A_{1\max} = \frac{kT}{m_1 m_2} * \frac{m_1 m_2}{k(m_1 + m_2)} * 2 = \frac{2T}{(m_1 + m_2)}$$

Thus, the effect of a finite stiffness is to increase the steady axial load on m_1 by a factor of 2. Note that this result does not depend on the spring stiffness or the relative magnitude of the two masses.

Three Mass Model of Ignition Shock

To understand this problem at its most fundamental level, consider a bare-bones model having only three massive elements, m_1, m_2 and m_3, each pair connected by a spring whose stiffness is k. At t = 0 the rocket is ignited and thrust T, acting as a step function is applied to m_3. The mass displacements are measured from the launch pad upward. Gravity is neglected since it is usually much smaller than thrust.



The three coupled equations of motion are:

$$m_1 \ddot{x}_1 = -k(x_1 - x_2)$$
$$m_2 \ddot{x}_2 = k(x_1 - x_2) - k(x_2 - x_3)$$
$$m_3 \ddot{x}_3 = k(x_2 - x_3) + T$$

Next, take the Laplace transform, assuming the stack starts from rest with no initial displacements.

$$m_1 s^2 \bar{x}_1 + k\bar{x}_1 = k\bar{x}_2$$
$$m_2 s^2 \bar{x}_2 + 2k\bar{x}_2 = k(\bar{x}_1 + \bar{x}_3)$$
$$m_3 s^2 \bar{x}_3 + k\bar{x}_3 = k\bar{x}_2 + \frac{T}{s}$$

Eliminate $\bar{x}_2$:

$$\bar{x}_2 = (m_1 s^2 + k)\bar{x}_1 / k \text{, and}$$

$$(m_2 s^2 + 2k)(m_1 s^2 + k)\bar{x}_1 = k^2(\bar{x}_1 + \bar{x}_3)$$

$$(m_3 s^2 + k)\bar{x}_3 = \frac{T}{s} + (m_1 s^2 + k)\bar{x}_1$$

Then, eliminate $\bar{x}_3$:

$$\bar{x}_3 = \frac{\frac{T}{s} + (m_1 s^2 + k)\bar{x}_1}{(m_3 s^2 + k)} \text{, and}$$

$$(m_2 s^2 + 2k)(m_1 s^2 + k)\bar{x}_1 - k^2 \bar{x}_1 = k^2 \frac{\frac{T}{s} + (m_1 s^2 + k)\bar{x}_1}{(m_3 s^2 + k)}$$

$$(m_1 m_2 s^4 + k(2m_1 + m_2)s^2 + k^2)\bar{x}_1 - k^2 \frac{m_1 s^2 + k}{m_3 s^2 + k}\bar{x}_1 = \frac{k^2 T}{s(m_3 s^2 + k)}$$

At this point it becomes expedient to aggregate the stack in a way makes all three masses the same. Then,

$$s^2(m^2 s^2 + 3km)\bar{x}_1 = \frac{k^2 T}{s(ms^2 + k)}$$

However, what we really want is the acceleration of the topmost (first) mass, A_1:

$$\bar{A}_1 = \frac{k^2 T}{s(ms^2 + k)(m^2 s^2 + 3km)}$$

Before inversion, rewrite this as

$$\frac{m^3 \bar{A}_1}{k^2 T} = \frac{m\bar{A}_1}{T} = \frac{1}{s} * \frac{1}{s^2 + \omega_n^2} * \frac{1}{s^2 + 3\omega_n^2}, \text{ where}$$

$$\omega_n^2 = \frac{k}{m}.$$

Expand this into partial fractions:

$$\frac{m\bar{A}_1}{T} = \frac{1/3}{s} - \frac{s/2}{s^2+\omega_n^2} + \frac{s/6}{s^2+3\omega_n^2}.$$

Next, invert the Laplace transform term by term:

$$\frac{mA_1(t)}{T} = \frac{1}{3} - \frac{1}{2}\cos(\omega_n t) + \frac{1}{6}\cos(3\omega_n t).$$

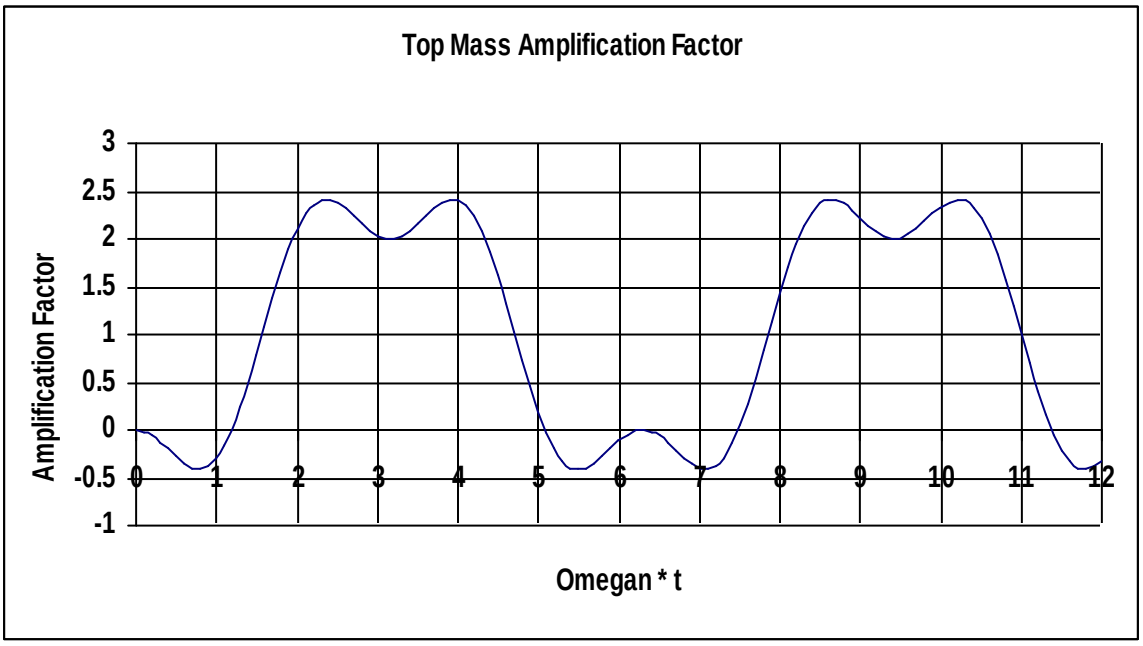
The steady state acceleration is

$$A_{ss} = \frac{T}{3m}$$

Then,

$$\frac{A_1(t)}{A_{ss}} = 1 - \frac{3}{2}\cos(\omega_n t) + \frac{1}{2}\cos(3\omega_n t)$$

This is plotted below as the Amplification Factor. It shows a maximum value of about 2.4142.



Next, find the acceleration of the second mass:

$$\bar{A}_2 = \frac{ms^2+k}{k}\bar{A}_1 = \frac{kT}{m^2 s(s^2+3\omega_n^2)} = \frac{\omega_n^2 T}{ms(s^2+3\omega_n^2)}.$$

The partial fraction expansion is

$$\frac{m\bar{A}_2}{T} = \frac{1/3}{s} - \frac{(1/3)s}{s^2 + 3\omega_n^2}$$

This can be inverted to give

$$\frac{mA_2(t)}{T} = \frac{1}{3} - \frac{1}{3}\cos\sqrt{3}\omega_n t$$

Normalizing by the steady state acceleration,

$$\frac{A_2}{A_{ss}} = (1 - \cos\sqrt{3}\omega_n t)$$

We see, by inspection, that the maximum value of this ratio is

$$\frac{A_2}{A_{ss}} = 2.$$

Finally, seek the maximum acceleration of the base element, x_3:

$$\bar{x}_3 = \frac{T}{s(ms^2 + k)} + \bar{x}_1, \text{ or}$$

$$\bar{A}_3 = \frac{sT}{ms^2 + k} + \bar{A}_1 = \frac{T}{m} * \frac{s}{s^2 + \omega_n^2} + \bar{A}_1$$

Then, since

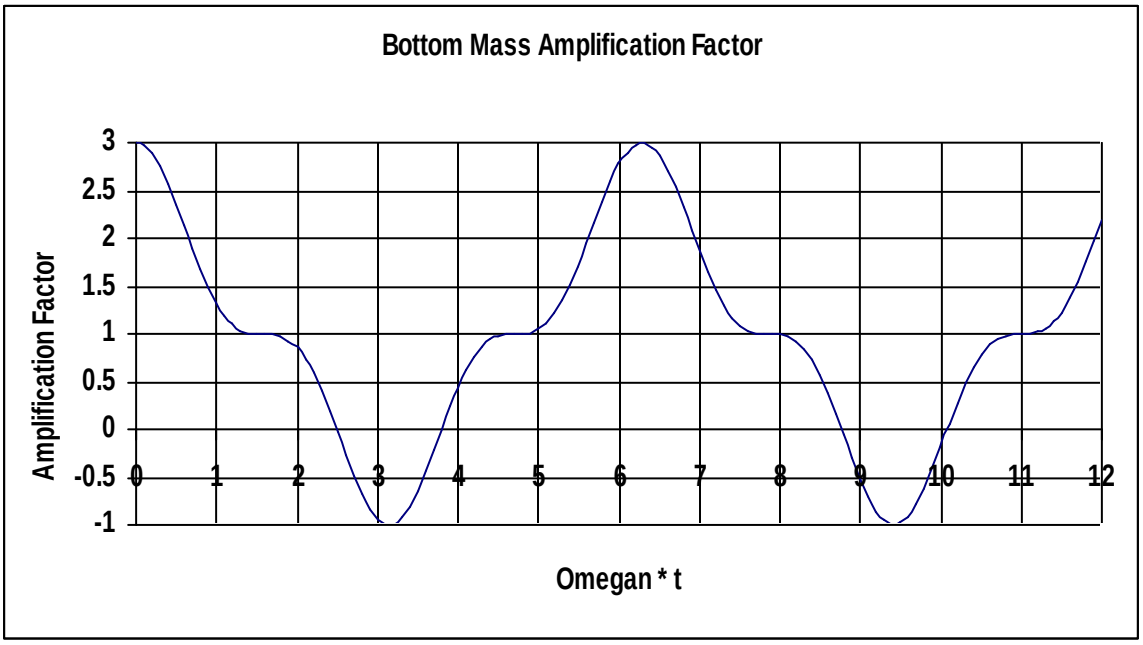
$$T = 3mA_{ss},$$

$$\bar{A}_3 = 3A_{ss}\frac{s}{s^2 + \omega_n^2} + \bar{A}_1$$

The inversion is easy:

$$\frac{A_3}{A_{ss}} = 3\cos(\omega_n t) + 1 - \frac{3}{2}\cos(\omega_n t) + \frac{1}{2}\cos(3\omega_n t) = 1 + \frac{3}{2}\cos(\omega_n t) + \frac{1}{2}\cos(3\omega_n t)$$

As shown in the plot below, the maximum Amplification Factor is 3.



Then, the amplification factors do not depend on element or total mass, nor do they depend on the stiffnesses between elements. In summary

Element	Amplification Factor
1 (top)	2.4142
2 (middle)	2
3 (bottom)	3